



Republic of Rwanda  
Ministry of Education



**RTB** | RWANDA  
TVET BOARD

**CSALS401**

**LED AND LCD SCREEN SET UP**

**Set Up LED and LCD Screen**

**Competence**

**RQF Level:** 4

**Learning Hours**

**Credits:** 7



**Sector:** ICT & MULTIMEDIA

**Trade:** COMPUTER SYSTEM AND ARCHITECTURE

**Module Type:** Specific

**Curriculum:** ICTCSA4001- TVET Certificate III in Computer System and  
Architecture

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|                                        |                                                                                                                                                                                                                                                                                                                                                           |            |  |                             |                             |            |                   |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--|-----------------------------|-----------------------------|------------|-------------------|
| <b>Purpose statement</b>               | This module describes the skills, knowledge and attitude required to Set Up LED and LCD Screen. This module is intended to prepare students pursuing TVET Level 4 in Computer System and Architecture. At the end of this module the students will be able to Assemble the LED/LCD, Configure LED/LCD Display Settings, Troubleshoot LED/LCD Screen Setup |            |  |                             |                             |            |                   |
| <b>Learning to be assumed in place</b> | <ul style="list-style-type: none"><li>• Digital Electronics</li><li>• Fundamentals of Electronics</li></ul>                                                                                                                                                                                                                                               |            |  |                             |                             |            |                   |
| <b>Delivery modality</b>               | <b>Training delivery</b>                                                                                                                                                                                                                                                                                                                                  |            |  | <b>100%</b>                 | <b>Assessment</b>           |            | <b>Total 100%</b> |
|                                        | <b>Theoretical content</b>                                                                                                                                                                                                                                                                                                                                |            |  | <b>30%</b>                  | <b>Formative assessment</b> | <b>30%</b> | <b>50%</b>        |
|                                        | <b>Practical work:</b>                                                                                                                                                                                                                                                                                                                                    |            |  | <b>70%</b>                  |                             | <b>70%</b> |                   |
|                                        | Group project and presentation                                                                                                                                                                                                                                                                                                                            | <b>30%</b> |  |                             |                             |            |                   |
|                                        | Individual project /Work                                                                                                                                                                                                                                                                                                                                  | <b>40%</b> |  |                             |                             |            |                   |
|                                        |                                                                                                                                                                                                                                                                                                                                                           |            |  | <b>Summative Assessment</b> |                             | <b>50%</b> |                   |

## Elements of Competency and Performance Criteria

| Elements of competency                         | Performance criteria                                                                                                            |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Assembling the LED/LCD panels</b>        | 1.1. LED/LCD Screen components are properly identified according to the manufacture's guide                                     |
|                                                | 1.2. LED/LCD screen components are properly handled based on manufacture's guideline and safety procedures                      |
|                                                | 1.3. LED/LCD screen components are correctly connected based on stable and reliable functionality of the LED/LCD Screen         |
| <b>2. Configuring LED/LCD Display Settings</b> | 2.1. LED/LCD Screen configuration software are identified based on the manufacture's guide                                      |
|                                                | 2.2. Visual parameters settings are configured based on required optimal display quality                                        |
|                                                | 2.3. Modes of loading content on LED/LCD Screen are identified based on user needs                                              |
|                                                | 2.4. Content is perfectly loaded on LED/LCD Screen based on manufacture's guide                                                 |
|                                                | 2.5. Functionality testing is correctly performed according to the intended use                                                 |
| <b>3. Troubleshooting LED/LCD Screen Setup</b> | 3.1. Dead pixels, color accuracy, and uniformity across the screen are effectively checked in accordance with user requirements |
|                                                | 3.2. Preventive maintenance measures are applied based on problem identified.                                                   |
|                                                | 3.3. Corrective maintenance measures are applied based on problem identified.                                                   |

### Intended Knowledge, Skills and Attitude

| Knowledge                                                                                                                                                                      | Skills                                                                                                                                                                                                                                                                                                                                                                                         | Attitude                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>✓ Description of LED and LCD</li> <li>✓ Identify LED/LCD Screens Components</li> <li>✓ Describe mounting techniques concepts</li> </ul> | <ul style="list-style-type: none"> <li>✓ Analysis of a LED/LCD Screen Display</li> <li>✓ Prepare installation site</li> <li>✓ Assemble of LED/LCD screen</li> <li>✓ Develop a LED Matrix Display</li> <li>✓ Configure of Visual Parameters</li> <li>✓ Perform preventive and corrective maintenance</li> <li>✓ Troubleshoot the LED/LCD Screens</li> <li>✓ Test the LED/LCD Screens</li> </ul> | <ul style="list-style-type: none"> <li>✓ Attentive</li> <li>✓ Problem-Solving</li> <li>✓ Teamwork</li> <li>✓ Adaptability</li> <li>✓ Time management</li> <li>✓ Integrity</li> </ul> |

### Course content

|                          |                                                                                                                                                                                                                                                   |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Learning outcomes</b> | <b>At the end of the module the learner will be able to:</b> <ol style="list-style-type: none"> <li>1. Assembling the LED/LCD panels</li> <li>2. Configuring LED/LCD Display Settings</li> <li>3. Troubleshooting LED/LCD Screen Setup</li> </ol> |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Learning outcome 1: Assembling the LED/LCD panels

**Learning hours: 25**

### Indicative content

- **Identification of LED and LCD screen components**

- ✓ Description of LED and LCD

-  Definition

-  Benefits

-  Application

-  Features

- ✓ Display technologies

- ✓ LED/LCD Screens Components

-  Display Panel

-  Video Input Interfaces

-  Data Cables

-  Power Cables

-  Backlighting System

-  Power supply

-  Filters

-  Controller and Driver Circuits

- **Handling Screen components**

- ✓ Prepare installation site.

-  Safety Precautions

-  Location Selection

-  Weather protection

-  Cable management

-  Electrical requirements

- ✓ Mounting Techniques

-  Wall

-  Ceiling

-  Curved

 Ground Stacking

- **Assembling of LED/LCD screen**

- ✓ Interfacing technologies

-  HDMI/DisplayPort

-  Wireless

-  Ethernet/IP Based

- ✓ LED/LCD Assembling techniques

-  Daisy Chain

-  Clockwise

-  Anti-Clockwise

-  Zigzag

- ✓ Tagging

- ✓ Test connectivity

### Resources required for the learning outcome

| Resources required for the learning outcome |                                                                                                                                                                                                       |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Equipment</b>                            | <ul style="list-style-type: none"><li>▪ LED panel</li><li>▪ LCD panel</li><li>▪ LCD Controller</li><li>▪ UPS</li><li>▪ Laptop Computer</li><li>▪ Surge Protector</li><li>▪ Drilling machine</li></ul> |
| <b>Materials</b>                            | <ul style="list-style-type: none"><li>▪ Power cables</li><li>▪ CAT6</li><li>▪ cable ties</li><li>▪ insulator</li><li>▪ RJ45</li><li>▪ Mounting Brackets</li><li>▪ Drill Bits</li></ul>                |

|                                            |                                                                                                                                                                                                                                           |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | <ul style="list-style-type: none"> <li>▪ Data cables</li> <li>▪ Screws</li> </ul>                                                                                                                                                         |
| <b>Tools</b>                               | <ul style="list-style-type: none"> <li>▪ Tape measure</li> <li>▪ Crimping tool</li> <li>▪ Wire stripper</li> <li>▪ Hammer</li> <li>▪ Pliers</li> <li>▪ Screwdrivers</li> <li>▪ ladder</li> <li>▪ Scaffold</li> <li>▪ ESD Tools</li> </ul> |
| <b>Facilitation techniques</b>             | <ul style="list-style-type: none"> <li>▪ Brainstorming</li> <li>▪ Demonstration and simulation</li> <li>▪ Individual and group work</li> <li>▪ Group discussion</li> <li>▪ Research</li> </ul>                                            |
| <b>Formative assessment methods /(CAT)</b> | <ul style="list-style-type: none"> <li>▪ Written assessment</li> <li>▪ Oral presentation</li> <li>▪ Performance assessment</li> </ul>                                                                                                     |

**Indicative content**

- **Develop LED Matrix Display**
  - ✓ Identify components.
    -  Microcontroller
    -  LED dot matrix modules
    -  Connecting Wires
    -  Controller board
    -  Power Supply
    -  Power connector
  - ✓ Interconnect display modules
  - ✓ Mount on circuit board
  - ✓ Program microcontroller
  - ✓ Test
- **Identification of configuration software**
  - ✓ Display Control software
  - ✓ Video processing software
  - ✓ Media Player software
  - ✓ Mapping software
  - ✓ Calibration software
- **Configuration of Visual Parameters**
  - ✓ Drivers installation
  - ✓ Calibration
    -  Refresh rate
    -  Overscan
    -  Underscan
  - ✓ Light
    -  Brightness
    -  Contrast



✓ Color

 Color temperature

 Saturation

 Vibrance

 Sharpness

 Gamma

 Profiles/Presets

✓ Size

 Resolution

 Aspect ratio

• **Perform content loading.**

✓ Ethernet

✓ HDMI/DP/DVI

✓ USB

✓ Wireless

✓ Video Switcher

✓ Live Feed and Broadcasting

✓ Media Players

• **Functionality testing**

✓ Electrical

 Power ON/OFF

 Cabling

✓ Data and display parameters

 Test Colors

 Test Patterns

 Brightness and Contrast Test

 Video Playback Test

 Pixel Mapping Test



Connectivity Test



Input/Source Test

### Resources required for the indicative content

|                                |                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Equipment</b>               | <ul style="list-style-type: none"> <li>▪ Computer,</li> <li>▪ Monitor,</li> <li>▪ Power Supply,</li> <li>▪ Switch,</li> <li>▪ Storage devices,</li> <li>▪ Scanning Control Board</li> </ul>                                                                                                                                                     |
| <b>Materials</b>               | <ul style="list-style-type: none"> <li>▪ Internet bundles</li> <li>▪ Connectors</li> <li>▪ Labelling tags</li> <li>▪ Power supply box</li> <li>▪ HDMI</li> <li>▪ DisplayPort</li> <li>▪ VGA Cables</li> <li>▪ jumper wires</li> <li>▪ Max7219 Dot Matrix LED</li> <li>▪ Arduino Uno</li> <li>▪ Breadboard</li> <li>▪ Power connector</li> </ul> |
| <b>Tools</b>                   | <ul style="list-style-type: none"> <li>▪ LED mPlayer</li> <li>▪ Controller Software</li> <li>▪ Content Management System Software</li> <li>▪ Special Video card</li> <li>▪ Multi-Function Card</li> </ul>                                                                                                                                       |
| <b>Facilitation techniques</b> | <ul style="list-style-type: none"> <li>▪ Demonstration and simulation</li> <li>▪ Individual and group work</li> <li>▪ Practical exercise</li> <li>▪ Individualized</li> </ul>                                                                                                                                                                   |

|                                                      |                                                                                                                                                                           |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                      | <ul style="list-style-type: none"> <li>▪ Trainer guided</li> <li>▪ Group discussion</li> </ul>                                                                            |
| <b>Formative assessment methods</b><br><b>/(CAT)</b> | <ul style="list-style-type: none"> <li>▪ Written assessment</li> <li>▪ Oral presentation</li> <li>▪ Performance assessment</li> <li>▪ Product based assessment</li> </ul> |

| <b>Learning outcome 3: Troubleshooting LED/LCD Screen Display</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Learning hours: 20</b> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <b>Indicative content</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                           |
| <ul style="list-style-type: none"> <li>• <b>Analysis of a LED/LCD Screen Display</b> <ul style="list-style-type: none"> <li>✓ Dead pixel inspection <ul style="list-style-type: none"> <li> Vertical or Horizontal lines</li> <li> Stuck pixels</li> </ul> </li> <li>✓ Checking accuracy <ul style="list-style-type: none"> <li> Color</li> <li> Uniformity</li> <li> Blurry or distorted images</li> </ul> </li> <li>✓ Common LCD/LED Screen issues <ul style="list-style-type: none"> <li> Backlight bleeding</li> <li> Screen flickering</li> <li> Image retention</li> <li> Power issues</li> <li> Overheating</li> </ul> </li> </ul> </li> <li>• <b>Perform preventive maintenance</b> <ul style="list-style-type: none"> <li>✓ Configuration Backup</li> <li>✓ Screen Cleaning <ul style="list-style-type: none"> <li> Proper cleaning tools</li> <li> Dust and dirt cleaning</li> </ul> </li> </ul> </li> </ul> |                           |

- ✓ Electricity
  -  Power fluctuation protection
  -  Secure connection
- ✓ Regular inspection
- ✓ Update firmware and drivers
- ✓ Screen Burn-in prevention
  -  Avoid high contrast images
  -  Use screensavers
  -  Regulate temperature
- **Perform corrective maintenance**
  - ✓ Identify diagnostic tools
    -  Software
    -  Hardware
  - ✓ Configuration restore
  - ✓ Calibration and Color Uniformity
  - ✓ Repair and replacement
  - ✓ Testing and Validation
  - ✓ Documentation

#### Resources required for the indicative content

|                                |                                                                                                                                                                                        |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Equipment</b>               | <ul style="list-style-type: none"> <li>▪ Computer</li> <li>▪ Monitor</li> <li>▪ Power Supply</li> <li>▪ Switch</li> <li>▪ Storage devices</li> <li>▪ Scanning Control Board</li> </ul> |
| <b>Materials</b>               | <ul style="list-style-type: none"> <li>▪ Internet bundles</li> </ul>                                                                                                                   |
| <b>Tools</b>                   | <ul style="list-style-type: none"> <li>▪ Web browser</li> <li>▪ Controller Software</li> </ul>                                                                                         |
| <b>Facilitation techniques</b> | <ul style="list-style-type: none"> <li>▪ Demonstration and simulation</li> </ul>                                                                                                       |

|                                            |                                                                                                                                                                                                               |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | <ul style="list-style-type: none"> <li>▪ Individual and group work</li> <li>▪ Practical exercise</li> <li>▪ Individualized</li> <li>▪ Trainer guided</li> <li>▪ Group discussion</li> </ul>                   |
| <b>Formative assessment methods /(CAT)</b> | <ul style="list-style-type: none"> <li>▪ Written assessment</li> <li>▪ Oral presentation</li> <li>▪ Performance assessment</li> <li>▪ Product based assessment</li> <li>▪ Project based assessment</li> </ul> |

### Integrated/Summative assessment

#### Integrated situation

ABBX Company Ltd is a private company located in Musanze District, it is specialized in Entertainment industry. ABBX Company Ltd in charge of all Audio-Visual activities in the Musanze District Trade Fair. However, the exhibitors play music at their stands, and this creates a problem of noise interference which can't let them pass any message via audio to the whole audience. The Organizers have requested the ABBX Company Ltd to help them solve this issue by using Visual solution. ABBX Company Ltd have suggested to solve the problem by setting up LED Screen Wall.

As a Computer System Integration technician, you are requested to set up LED Screen Wall System which helps organizers to present their messages in any visual format. All required components are provided.

The system will have the way of displaying videos, pictures, live feeds, and standing at 2 meters from ground and having a capacity of displaying 5x4 meters image.

After installation, the technician must operate the system and report the work done to the employer.

This work is supposed to be done in 5 hours.

#### Resources

|              |                                                                                                                                                                 |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tools</b> | <ul style="list-style-type: none"> <li>▪ Crimping tool, Wire stripper, Drilling machine, Hammer, Ladder, Pliers, Screw drivers, Controller software,</li> </ul> |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                             | Calibration software                                                                                                                                                                          |                                                 |             |    |                  |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|-------------|----|------------------|
| <b>Equipment</b>                            | <ul style="list-style-type: none"> <li>Computer, Power Supply, Switcher, Storage devices, Scanning Control Board</li> </ul>                                                                   |                                                 |             |    |                  |
| <b>Materials/ Consumables</b>               | <ul style="list-style-type: none"> <li>Internet bundles, Connectors, Cable ties, trunks, Pipes, Screws, Wall Mounts, Power extension, Labelling tags, Power supply box, LED panels</li> </ul> |                                                 |             |    |                  |
| Assessable outcomes                         | Assessment criteria (Based on performance criteria)                                                                                                                                           | Indicator                                       | Observation |    | Marks allocation |
|                                             |                                                                                                                                                                                               |                                                 | Yes         | No |                  |
| <b>Assembling the LED/LCD panels (30%)</b>  | Identification of LED/LCD Screen components                                                                                                                                                   | LED Technology is used                          |             |    | 3                |
|                                             |                                                                                                                                                                                               | LED components are selected                     |             |    | 4                |
|                                             | Handling LED/LCD screen components                                                                                                                                                            | Installation site is prepared                   |             |    | 5                |
|                                             |                                                                                                                                                                                               | Mounting techniques are applied                 |             |    | 4                |
|                                             | Connection of LED/LCD screen components                                                                                                                                                       | Connection and structure techniques are applied |             |    | 5                |
|                                             |                                                                                                                                                                                               | Assembling structure techniques are applied     |             |    | 6                |
|                                             |                                                                                                                                                                                               | Connection cables are tagged                    |             |    | 5                |
| <b>Configuring LED/LCD Display Settings</b> | Identification of LED/LCD Screen configuration software                                                                                                                                       | Display Control software is selected            |             |    | 4                |
|                                             |                                                                                                                                                                                               | Mapping software is selected                    |             |    | 3                |
|                                             |                                                                                                                                                                                               | Calibration is selected                         |             |    | 4                |

|                                            |                                                                           |                                               |     |   |   |
|--------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------|-----|---|---|
| (40%)                                      | Configuration of Visual parameters settings                               | Drivers are installed.                        |     |   | 4 |
|                                            |                                                                           | Color and calibration settings are configured |     |   | 4 |
|                                            |                                                                           | Resolution and Aspect Ratio are configured    |     |   | 5 |
|                                            |                                                                           | Contrast and Brightness are configured        |     |   | 4 |
|                                            | Loading Content on LED/LCD Screen                                         | Ethernet mode is used                         |     |   | 4 |
|                                            |                                                                           | Video Switcher mode is used                   |     |   | 4 |
|                                            |                                                                           | Live Feed and Broadcasting mode is used       |     |   | 6 |
|                                            | Perform functionality testing                                             | Power test is performed                       |     |   | 4 |
| Data based tests are performed             |                                                                           |                                               |     | 4 |   |
| Troubleshooting LED/LCD Screen Setup (30%) | Checking of Dead pixels, color accuracy, and uniformity across the screen | Accuracy is checked.                          |     |   | 4 |
|                                            |                                                                           | Dead pixels are inspected.                    |     |   | 4 |
|                                            | Application of preventive maintenance measures                            | Configuration Backup is performed             |     |   | 3 |
|                                            |                                                                           | Power fluctuation protections are installed   |     |   | 4 |
|                                            |                                                                           | Screen Burn-in prevented                      |     |   | 3 |
|                                            | Total marks                                                               |                                               | 100 |   |   |
| Percentage Weightage                       |                                                                           | 100%                                          |     |   |   |

## References:

1. Brown, A. (2020). The Complete Guide to LED Screen Installation. New York, NY: Tech Publications.
2. Johnson, M. R. (2019). A Practical Approach to LCD Screen Setup: Best Practices for Installation and Calibration. Chicago, IL: Tech Experts Press.
3. Smith, J. P. (2018). LED Video Walls: Design, Installation, and Troubleshooting. Los Angeles, CA: Media Solutions Inc.
4. White, E. L. (2017). Advanced LCD Display Techniques for Professionals. Boston, MA: Graphics Press.
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10. Richardson, S. M. (2011). LED and LCD Video Walls: Installation Techniques for Professional Audio-visual Installers. Seattle, WA: AV Press.